**Radiography as Women's Work: British Nurses and Diagnostic X-Rays in the First World War**

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"*The work of an X-ray nurse is very interesting and grows rapidly in scope, for women can in many cases take the place of the male assistants now employed as operators… Day by day, especially during this time of war and stress, X-ray work increases in its value and importance. No hospital can do without it…*"[1]

[1] An X-Ray Worker. 'Concerning X-ray Work', *The Nursing Times*, Dec. 25 1915, 11(556), pp. 1599-1600. Copies of *The Nursing Times* available online at https://archive.org/details/pub_nursing-times?tab=collection.

Within this essay, I will analyse the underexamined role of British nurses in radiography during the First World War. Firstly, I will discuss wartime nursing as gendered work within contexts of military and medical reform. Secondly, I will examine relations between wartime, nursing, and radiography. The novel status of radiology as a field, and its subsequently unregulated nature in Britain at the time, combined with the demands for military radiography, provided opportunities for women to commandeer X-ray technologies. Nurses in wartime were therefore able to create and access new opportunities in radiography, shaping the boundaries of their profession. However, this was a phenomenon in flux, and early female pioneers within the field both created and destroyed barriers for other women, on the basis of class and divisions between volunteer and professional nurses. Finally, I will discuss the varieties of radiological nursing practice within Britain and abroad in Allied military hospitals during the First World War.

The contributions of assistants, technicians and nurses, often women, are frequently marginalised in historical narratives.[2] While historical work has been done recognising the role of women in British radiotherapy,[3] historiographies relating to radioactivity have been mainly occupied with accounts of significant discoveries made in the field, or biographies of the figures who made these discoveries.[4] This has led me to investigate the roles of nurses as innovators[5] within radiographical practice in the period where diagnostic X-rays were secured as a significant, necessary aspect of battlefield medicine.[6] Nursing histories can be difficult to reconstruct, as their subjects tend not to record their experiences or contributions to medical practice.[7] However, nursing journals exist as spaces for nurses to share knowledge and

[2] Hallett, 2009, xi. Jones, Martin and Wolf, 2021, 4-7.
[3] Elston, 2001, 90. Moscucci, 2007, 140. Murphy, 1986, Chap. 5.
[4] Rentetzi, 2007, xiii.
[5] Hallett, 2013, 99.
[6] Banerjee and Thomas, 2013, 47-48. Cirillo, 2000, 1238-1239.
[7] Buresh and Gordon, 2000, 4. Hallett, 2009, xi. Long, 2013, 50.

experiences, making them a very valuable resource as relatively rare examples of detailed nursing records.[8] Therefore, I have chosen to use *The Nursing Times* in creating this account, which Hallett identifies as one of the three most significant journals in Britain during the 20th Century.[9] Receptions towards such journals within nursing communities are mixed. As frontline nurse M.M. James wrote in her diary in 1914, ‘‘our comic papers now are the *Nursing Mirror* and *[Nursing] Times.* We read them aloud at meals. They do so slop over with sickly sentiment about the ‘brave women at the front’ and so on…’.[10] However, the longevity and dissemination of this journal suggest it was a widely appreciated resource. It should also be noted that the nursing literature of the period sometimes refers to X-ray images as ‘radiograms’ or ‘radiographs’. For example, see the highly influential *A Text Book of War Nursing* by British frontline nurse Violetta Thurstan, described by Hallett as ‘what is probably the most significant and comprehensive contemporary text on wartime nursing practice’, and therefore an important source for this essay.[11]

## Section I - Wartime Nursing as Gendered Work

From the inception of military medicine in the British armed forces around the 17th Century,[12] and until the mid-19th Century, most military nurses were men.[13] The Crimean war in 1853-1856, however, exposed significant issues with medical practice in the British military, catalysing campaigns for the rights of British women to work as nurses within war zones.[14] Further influenced by wider movements to regulate and professionalise medicine in Britain,[15]

[8] Brooks and Hallett, 2015, 7.
[9] Hallett, 2013, 96.
[10] Diary of Miss M.M. James, 23 December 1914, Peter Liddle 1914–1918 Personal Experience Archive, University of Leeds, Women Collection. See Watson, 2002, 494.
[11] Hallett, 2009, 13-14. Quotation from Hallett, 2013, 108.
[12] Meyer, 2019, 28.
[13] Vuic, 2013, 22-24.
[14] Ibid., 22-23, 42-43. Gill, 2014, 5.
[15] Meyer, 2019, 27, 43.

the subsequent disturbances were settled only by the end of the 19th Century after significant public debate.[16] During the Boer War of 1899-1902, more men than women were still employed as nurses within the British military,[17] but the gender dynamics of nursing had begun to shift.[18] Firstly, the increase of women within military nursing was necessitated by both an increase in demand and an absence of men during wartime.[19] This occurred further within the First World War, a period widely considered to have presented new opportunities for women in Britain.[20] By this time, being a nurse was generally considered a respectable and even admirable profession.[21] Secondly, these shifts involved feminisation of the nursing profession.[22] This process involved arguments that nursing was best suited by women as nurturing, natural caregivers,[23] with the effect of both increasing numbers of women in military nursing and excluding men.[24] These arguments posited a complex issue for feminist agendas in medicine.[25] While successful in increasing the number of female nurses, and providing them with professional opportunities, they could be problematic for women who wanted to be doctors but were pigeonholed as nurses instead.[26] Indeed, the feminisation of nursing implied their value was due to gender rather than medical competency.[27] During wartime, romanticisation of nursing as patriotic, feminine caregiving further increased societal acceptance of women nurses in the military.[28] Finally, nursing was undergoing changes to professionalise and elevate it into a '[profession] suitable for respectable women'.[29] Part of nursing reform was class-based as

[16] Ibid., 22-23, 42.
[17] Summers, 1988, 1-2.
[18] Brooks and Hallett, 2015, 2.
[19] Ibid., 9-10. Hallett, 2009, 7.
[20] Watson, 2002, 485.
[21] Fara, 2018, 27-29.
[22] Meyer, 2019, 37.
[23] Ibid, 37-38. Heggie, 2015, 271-272.
[24] Meyer, 2019, 42-43, 48, 52.
[25] Ibid, 33. Heggie, 2015, 268, 273.
[26] Ibid.
[27] Brooks and Hallett, 2015, 2.
[28] Fara, 2018, 28, 198, 216.
[29] Heggie, 2015, 274.

well as gender-based:[30] Hacker describes the goals of nursing reform as an attempt to move from a 'menial lower-class occupation to a respectable middle-class calling'.[31] This accords with the feminisation of nursing practice, [32] as the nurturing traits argued to be inherently feminine were associated strongly with the middle- and upper-classes.[33] Professionalisation functioned at different levels, however, and involved both improving the education and standards of the paid nurse and regulating the work of poorly trained upper-class 'lady volunteers'.[34] Elevating nursing entailed the promotion of scientific knowledge and technical skills. The new and deadly technologies during the War meant nurses had to assume significant responsibilities in caring for the wounded, developing new treatments and methodologies to cope with the novel horrors of war.[35] As Thurstan wrote in her influential book, 'in many ways nursing was more interesting in the early days of the war, when everything had to be improvised or adapted, than later on… how many lives and limbs were saved by the nurses' ready inventiveness and clever fingers?'.[36] The figure of the British nurse in the First World War, therefore, represents both challenging and embracing societal gender roles of the period, as an innovator within a caregiving profession.

Within the First World War there was a complex interplay between volunteer and professional nurses,[37] and significant tensions between them.[38] As the War progressed, the existence of (mainly middle- and upper-class) volunteer nurses with varying levels of training, and their increasing involvement in military hospitals, somewhat undermined the claims of professional nurses as to their status of their profession.[39] However, the presence of women in large numbers

[30] Meyer, 2019, 27.
[31] Hacker, 2012, 14 (in Meyer, 2019, 32). Hallett, 2009, 7.
[32] Heggie, 2015, 271.
[33] Ibid., 272.
[34] Ibid., 274.
[35] Brooks and Hallett, 2015, 5, 7.
[36] Thurstan, 1917, 11 (in Hallett, 2009, 224).
[37] Hallett, 2009, 10. Meyer, 2019, 34-38, 48-52. Palmer, 2015, 154.
[38] Watson, 2002, 486, 499-508.
[39] Ibid., 494, 499-500. Gill, 2013, 11 (in Meyer, 2019, 23). Hallett, 2009, 92.

within wartime allowed for a greater statement of female competency. Watson further draws our attention to the fact that volunteer nurses were 'untainted by professionalism', meaning that their role in the war was more socially acceptable than that of the professional nurse, appearing to be more altruistically caregiving.[40] As I will discuss in Section II, volunteers could be instrumental in creating networks and opportunities for other women.

Brooks and Hallett have identified three main incentives for British nurses working abroad during the First World War, which I will address in turn: humanitarianism, patriotism, and prestige.[41] The first is perhaps the most obvious,[42] given the appeal of nursing when romanticised as self-sacrificing and heroic.[43] It also leads to an interesting interplay with the second motivation: nurses' patriotic contributions in war were through healing medical practice rather than death.[44] This second motivation was highly influential.[45] Thurstan, for example, implied the patriotic intentions of her fellow professional nurses when she wrote '[t]he very natural desire of all nursing sisters who go abroad is to be sent as near as possible to the front… their only object is to be of real use'.[46] The word 'natural' demonstrates how common this sentiment must have been, as Thurstan considered this point to be almost obvious. Looking instead at volunteer nurses, Lady Parsons, an engineer influential particularly in the wartime period,[47] described the eagerness of upper- and middle-class women to volunteer as nurses, 'if only they might have something useful to do'.[48] The third motivation, prestige, is very significant in understanding the roles of nurses in the war. As Brooks and Hallett write, war was 'an opportunity to demonstrate nursing expertise'.[49] Therefore, alongside the gendered work of

[40] Watson, 1997, 51.
[41] Brooks and Hallett, 2015, 4.
[42] Hallett, 2009, 178.
[43] Ibid., 39.
[44] Ibid., 6, 8. Meyer, 2019, 25-26.
[45] Hallett, 2015, 93-94. Watson, 1997, 32.
[46] Thurstan, 1917, 22.
[47] Pursell, 1993, 80-81.
[48] Parsons, 1916, 229-230.
[49] Brooks and Hallett, 2015, 4.

caregiving, nurses could utilise technologies such as X-rays in wartime to gain greater status within medicine.[50] This is consistent with efforts to regulate and elevate nursing within the late 19th and early 20th Century.[51] However, professionalisation within medicine also had the effect of creating public disdain in Britain towards nurses who sought to elevate their professional status.[52] There is a tension here, therefore, between efforts of nurses to attribute prestige to their profession, and public animosity towards women attempting to do so. However, the humanitarian and patriotic aspects of war work helped to alleviate this tension, as nurses could refer to these aims to gain opportunities and respect within public discourse.

## Section II - Wartime Nursing and Radiography

Having discussed the context of British military nursing before and during the First World War, I will now turn to radiographical practice. Within four months of the discovery of X-rays by W.C. Roentgen in 1895, they were being used to locate bullets and needles within the body and to diagnose fractures,[53] with early applications in battlefield medicine.[54] However, unwieldy, complex equipment and a lack of trained personnel limited their early medical use,[55] until the casualties of the First World War necessitated their widespread, crucial utilisation.[56] The discovery of X-rays also came, as previously discussed, within a time of intense British military and medical reform. Similarly, there was debate before the War as to what radiological work should be carried out by doctors, nurses, physicists, or specialists in X-rays.[57] Radiography would generally not have been performed by nurses in Britain at this point. Consider the following quotation from doctor Agnes Savill in *The Nursing Times*, from March 1914: '[t]he

[50] Ibid., 14-15. Watson, 2002, 486.
[51] Meyer, 2019, 33.
[52] Watson, 2002, 490.
[53] Moscucci, 2016, 148.
[54] Badash, 2003, 42. Banerjee and Thomas, 2013, 38-41.
[55] Curie, 1938, 297-298.
[56] Banerjee and Thomas, 2013, 47-48. Cirillo, 2000, 1238-1239. Hallett, 2009, 98.
[57] Moscucci, 2016, 152-153, 159-160, 180. Rentetzi, 2007, 19-20.

responsibility incurred in $x$-ray (sic) work is so great that it is usually undertaken only by the medical attendant… With long practice an intelligent nurse can recognise the methods of safe dosage in treatment with perfect accuracy, and she may thus become an invaluable assistant to a qualified physician. But if she is therefore tempted to administer $x$-rays on her own account… sooner or later the inevitable error, common to humanity, will occur, and she may be involved in a lawsuit disastrous to her career… The nurse does very little beyond keeping the instruments clean'.[58] This quotation illuminates the boundaries being drawn within British radiology at the time. The emphasis on how, even for her 'little' involvement, the nurse must be especially 'intelligent' and have had 'long practice', suggests Savill's suspicion towards the capabilities of nurses. It is also interesting to note that a female doctor wrote this passage. The quotation here suggests a desire for the female physician to distance herself from 'the nurse', perhaps as part of the struggle over women's places in medicine, and who could access and control medical X-ray technology.[59] Indeed, Savill worked within radiology,[60] and during the First World War was a pioneering radiologist working with diagnostic X-rays at a combat hospital in France.[61] The absence of men in military hospitals during the War, however, meant that skilled women nurses with access to radiographical training were well-positioned to work as administrators of this technology.[62] Combined with efforts in Europe to promote and regulate nursing, this provided opportunities for innovative British nurses to adopt radiographical work within their own sphere, and prove their professional competency.[63]

To provide further context to radiography as performed by British nurses in wartime, I will now discuss the work of female physicists in establishing radiology in Europe. Marie Curie, her

[58] Thomas, 2022, 318.
[59] See Heggie, 2015, for an extended discussion of the tensions between female doctors and nurses.
[60] Thomas, 2022, 211.
[61] Creamer, 2022.
[62] Brooks and Hallett, 2015, 10.
[63] Kelly, 2021, 542.

daughter Irène Joliot-Curie, and Lise Meitner, arguably the most famous female scientists of this period, all worked as physicists within the field of radioactivity (to become nuclear physics). They also all worked as radiology nurses with the Red Cross during World War One: the Curies in France, and Meitner for the Austrian army on the Eastern and Italian fronts.[64] Interestingly, this aspect of their identities as nurses is not widely known or analysed. Like many women, Curie registered with the medical service in France at the outbreak of war, despite no prior medical training.[65] Curie thus discovered that, troublingly, hospitals at and around the front did not have X-ray equipment. [66] Meitner similarly had 'harrowing' experiences nursing with inadequate radiographical equipment.[67] Curie's major impact on the use of radiology in Allied military hospitals is in part quantifiable, and she herself described it in a short book she wrote of her wartime experiences called *La Radiologie et La Guerre* – Radiology and War. Curie identified the following problems with radiology in France at this time: general ignorance, few facilities, few specialist staff (who were largely ignored), and isolated or ineffective attempts to spread the use of X-rays.[68] Therefore, she organised funds, equipment and personnel for the use of diagnostic X-rays in frontline hospitals, and invented the 'radiological car', a vehicle fitted with portable X-ray apparatus.[69] By the end of 1918, working often with her daughter, she had established around 500 radiological stations and 300 mobile X-ray machines.[70]

As well as improving radiographical technologies that British nurses overseas could access within the war, the Curies and Meitner acted as role models.[71] Curie in particular would have

[64] Curie, 1938, 292. Meitner, 1964, 10. Red Cross memorabilia from the Meitner Archives at Churchill College, Cambridge.
[65] Fara, 2018, 216.
[66] Curie, 1938, 282.
[67] Frisch, 1970, 407.
[68] Curie, 1921, 115, 119.
[69] Banerjee and Thomas, 2013, 45. Curie, 1938, 283.
[70] Curie, 1921, 118.
[71] Fara, 2018, 222-223. On the importance of female role models within radiology, see Rentetzi, 2007, 65-66, and Rayner-Canhams, 1997, 18.

been a familiar name for British nurses, often referenced in nursing journals as the discoverer of radium,[72] and she directly interacted with British nurses in wartime. For example, a passage in *The Nursing Times* in 1915 alludes to Curie's work in a Belgian military hospital with a staff of 21 British nurses: '[t]he famous French scientist, Mme. Curie, is doing $x$-ray [sic] work at the hospital, assisted by her daughter'.[73] In 1916, Curie ran the radiography department at the Edith Cavell Hospital in Paris, a memorial hospital for a British nurse executed in 1915 for helping Allied soldiers to escape occupied Belgium.[74] The hospital also functioned as a radiographical training school for professional nurses.[75] The reporting of Curie's work at this hospital in *The Nursing Times* is important in suggesting her influence on British women in radiography, as Cavell was also a significant inspirational figure for British nurses at the time.[76] The article also conveyed that this hospital was intended as a training school for nurses after the War, suggesting this would be a training opportunity available to British women.[77] Curie similarly established a training programme in radiology at the Radium Institute in Paris.[78] 150 women, 'recruited from all classes', were trained here by Curie during the war.[79] The comment on class, from Ève Curie's biography of her mother, indicates how Curie's programme was novel in providing opportunities across class and gender divides. Marie Curie also asserted her opinion within *La Radiologie et La Guerre* that women of 'average education' were capable of working as radiology technicians.[80] Curie's book would have been read in Britain after the war, making this an important statement from an influential figure as to the abilities of female radiographers. Her influence is reflected within a 1923 review of this book in the prestigious scientific journal

[72] E.g. 'Woman's Wider World', *The Nursing Times*. November 24 1906, 2(82), p. 1000. September 14 1907, 3(124), p. 809. March 27 1909, 5(204), p. 247.
[73] *The Nursing Times*, January 16 1915, 11(507), pp. 58-60.
[74] 'The Paris Memorial', *The Nursing Times*, October 21 1916, 12(599), p. 1232. Photograph of Curie with female students available from the National Library of Medicine, http://resource.nlm.nih.gov/101691059. Hoehling, 1957, 1320.
[75] Curie, 1921, 110.
[76] Hoehling, 1957, 1321.
[77] 'The Paris Memorial', *The Nursing Times*, October 21 1916, 12(599), p. 1232.
[78] Badash, 2003, 43. Curie, 1921, 110.
[79] Curie, 1921, 111-112. Quotation from Curie, 1938, 294.
[80] Curie, 1921, 112. Self-translated.

*Nature*, describing Curie as 'the distinguished author' and suggesting that 'an account of the British radiological activities during the War should be put on record'.[81] Therefore, Curie's work in wartime presented new technologies, opportunities, and arguments of competence for British nurses via radiography.

To examine shifts in British attitudes towards women radiographers, I will now discuss Lady Helena Gleichen. An upper-class woman, cousin to King George V,[82] Gleichen had the means to work as a volunteer in wartime, and the connections and abilities to create points of entry for other women in radiographic nursing. After completing radiology training in Paris and London with her friend Nina Hollings, the British War Office declined their offer of work: 'as no women had ever been known to be radiographers, we could not be employed as such'.[83] Gleichen therefore used money raised by her family to buy a portable X-ray unit.[84] Gleichen and Hollings worked thus in France for ten months, and on the Italian Front for two years.[85] Several women, such as volunteer nurse Vera Woodroffe, were employed by Gleichen and Hollings as radiography assistants.[86] This demonstrates how Gleichen's work created new opportunities for women, although the extent to which this impacted professional nursing remains unclear. Gleichen was also able to positively impact attitudes towards women radiographers within the Allied military. For example, she related in her autobiography how at the beginning of her stay in Italy, one of their radiographs was doubted and a man sent for to double check their work. Eventually, they would be called upon to check his work.[87]

[81] G.W.C.K., 'La Radiologie et la guerre. By Mme. P. Curie.', *Nature*, September 22 1923, 112(2812), p. 433.
[82] Fara, 2018, 220.
[83] Gleichen, 1940, 126 (in Fara, 2018, 220). Similarly, sisters Florence Ada Stoney and Edith Stoney applied to work abroad as radiologists, with the former having considerable experience in medical radiology and the latter being a physicist, but were refused by the War Office. See Fara, 2018, 229-230, Banerjee and Thomas, 2013, 49-54, and Thomas, 209.
[84] Gleichen, 1940, 126.
[85] *Journal of the Royal Society of Arts*, February 14 1947, 95(4737), p. 219.
[86] Gleichen, 1940, 262-263. Information on Vera Woodroffe from the Imperial War Museum Online Collections, catalogue numbers 9603-14 and Documents.3262. Medal card from the National Archives, Kew, Reference Number WO 372/23/45997.
[87] Gleichen, 1940, 176.

Interestingly, one reason that Gleichen gave for her success was that 'we attempt anything as we have no reputation to keep up, and these professors won't risk doing bad radiographs because they have their good name to think of'.[88] Their roles as volunteers in this case was advantageous, allowing them to take risks that professional nurses may not have taken for fear of harming their reputations, given aspirations regarding the elevation of nursing as discussed in Section I. How professional nurses may have felt about this is unclear. Plausibly, some may have had concerns about such attitudes within their profession, and how this would reflect upon themselves. However, Gleichen was able to utilise her class and position to create further opportunities for women in wartime and demonstrate to high-powered officials in the military and the medical profession that women could be successful radiographers. Gleichen received the Order of the British Empire for her war work,[89] starkly contrasting with the refusal of her services from the British War Office in the early war period and demonstrating a notable recognition of women's competency for radiological work within Britain.

Radiography during wartime also had links with women's rights movements, in part due to Marie Curie's role as a pioneer in the study of radioactivity.[90] Feminist activity included fundraising for X-ray equipment: it was reported in the Suffragist publication *The Common Cause* that the London Society for Women's Suffrage had raised over £1000 (around £60,000 in today's money) to send an X-ray ambulance to the front.[91] Similarly, *The Nursing Times* reported appeals by female nurses 'Miss A. Maud Fletcher' and 'Miss de Laveleye' to raise £200 (£12000) to take X-ray equipment to a Belgian frontline hospital.[92] Suffragists were also involved in organising medical teams abroad; *The Nursing Times* mentions two overseas units

[88] Ibid., 218.
[89] 'Lady Helena Gleichen', *The New York Times*, January 29 1947, p. 25.
[90] Moscucci, 2016, 148, 165-166, 281. Rentetzi, 2007, 27-28.
[91] Iglikowski-Broad, 2015.
[92] 'Guys' Nurses at Ghent', *The Nursing Times*, January 9 1915, 11(506), p. 46.

organised by suffragist Mabel Stobart (an upper-class woman)[93] entirely staffed by women, with specific reference to the X-ray facilities available.[94] Women thus created new opportunities for other women in wartime radiography.

## Section III - Varieties and Details of Practice

The inherently novel and interdisciplinary nature of radiology in the early 20th Century was a significant factor in how British nurses accessed X-ray technology. [95] As it had not yet been resolved who would be responsible for this work – doctors, nurses, physicists, or radiology specialists and so forth – radiography was not automatically and rigidly determined as men's work, allowing women to make their own claims. The important wartime model of interdisciplinary teams for medical practice was another factor for the success of nurses within radiographical practice.[96] Radiology as a specialist discipline, with multiple components such as the running of technology and the interpretation of images, suggests a need for a collaborative approach. This created opportunities for nurses to occupy some of these radiological roles within their medical teams. Moscucci draws attention to the resultant successes of 'teamwork' approaches to radiology treatment in France,[97] and the (female-staffed) Marie Curie Hospital in London after the War.[98] British nurses in wartime could therefore assert themselves in a new field which had not yet been earmarked as a male dominion.[99] Via radiography, nurses could articulate a new vision of the nursing profession.

[93] Blodgett, 2004.
[94] 'At The Chateau Tourlaville', *The Nursing Times*, February 13 1915, 11(511), pp. 189-190. 'Mrs. Stobart's Unit', *The Nursing Times*, March 20 1915, 11(516), p. 344. Florence Stoney was involved in radiography work at both of these institutions (Banerjee and Thomas, 2013, 52-53).
[95] Rayner-Canhams, 1997, 18, 1998, 94. Rentetzi, 2007, 36-37, 99.
[96] Hallett, 2009, 94.
[97] Moscucci, 2016, 170. Pinell, 2003, Chapter 4.
[98] Moscucci, 2016, 170.
[99] Fara, 2018, 267. Jones, Martin and Wolf, 2021, 4. Moscucci, 2016, 161-162. Rentetzi, 8, 85, 101.

The non-standardised nature of radiological practice meant a large variety in nurses' work, which complicates the presentation of a comprehensive picture of what this work entailed. I have roughly utilised the concept of a 'radiographical nurse', but this phrase encompasses a variety of practice, from specialist roles as radiography assistants to professional nurses who did radiography as another element of their work. In light of this, I will use a 1915 article in *The Nursing* Times credited only to 'an X-ray worker' as a primary account to illuminate the nature of their work.[100] The dissemination of this journal amongst British nurses, as previously discussed, suggests that this article would have been a relevant and accurate representation of the experiences of many nurses. Indeed, another issue of this journal directs its readers interested in X-ray work to this article.[101] The article firstly describes radiographical aspects of this work: enlarging, reducing, making photographs, giving the correct exposures, and keeping the X-ray tubes in order. The account mentions a radiographic resident who directs the work of the department, but also that the nurses are under the direction of a 'sister', who 'in smaller institutions does the work for which a resident would otherwise be responsible… she will often have to interpret the radiograms (the ignorance many doctors have of them is surprising) and see that they are properly taken by her staff.' Here we find evidence that nurses were engaged significantly in radiographical work, as well as that some nurses would have been undertaking *radiological* work, with the suggestion that some nurses would know more than the doctors they were working with about this X-ray technology.

One way to recognise the new opportunities for British nurses in military radiography is via adverts in *The Nursing Times* on how to become an 'efficient nurse' suitable for aiding the war effort, as well as for books on nursing training, which make explicit mention of X-rays.[102] While

[100] An X-Ray Worker, 'Concerning X-ray Work', *The Nursing Times*, December 25 1915, 11(556), pp. 1599-1600.
[101] 'Answers To Correspondents: X-ray Work (Peter Pan)', *The Nursing Times*, November 25 1916, p. 1420.
[102] *The Nursing Times*. May 23 1914, 10(473), p. 676. October 2 1914, 10(492), p. 1233. February 13 1915, 11(511), p. 181. December 4 1915, 11(553), p. 1510. September 8 1917, 13(645), p. 1054.

perhaps not to the same degree, more opportunities were also being created within domestic nursing in Britain, particularly as more men enlisted. For example, a Mr C.J.S. Thompson is credited with having arranged 'short courses of instruction for nurse assistants in radiography' at the Middlesex Hospital in 1916, carried out by a Mr. C.R.C. Lyster and Miss Marian Butler.[103] Similarly, in the Italian Hospital in Queen's Square, London, "[i]n the X-ray [sic] department, now that the assistant is on active service, the sisters are becoming experts in developing the plates."[104] Therefore, by the midpoint of the war, more opportunities both to teach and learn were becoming available to British nurses. However, they were not always able to work as radiographers within the British military. For example, Gleichen recounted how in 1916 her brother wrote to her that 'he wished to goodness the English authorities would allow [them] to come to his division as they have only one apparatus and that a rotten one, but that being women he supposed [they] should never be allowed'.[105] British women were usually allowed only as far as casualty clearing stations behind the front lines, which Thurstan ascribes to a 'chivalrous' British attitude.[106] This shows the variation in attitudes to women doing this work within Britain and the Allied military. Nevertheless, compared with Savill's account of the work British nurses were doing with radiography before the war as discussed in Section II, many nurses became able to adopt significantly more responsibilities within this field.

## Conclusion

Within this essay, I have analysed the roles of British nurses within radiography in the First World War, utilising a nursing journal and textbook as primary sources. In the first of three sections, I established a context of military and medical reforms within Britain, as well as of the

[103] 'Radiography', *The Nursing Times*, October 21 1916, 12(599), p. 1232.
[104] 'The Italian Hospital, Queens Square', *The Nursing Times*, July 10 1915, 11(532), p. 852.
[105] Gleichen, 1940, 192.
[106] Hallett, 2009, 15. Thomas, 2022, 209. Thurstan, 1917, 22. Watson, 2002, 505.

professionalisation of nursing and societal responses to it. In particular, this involved looking at some of the significant relations between volunteer and professional nurses within the relevant period, and also the various motivations of female nurses in wartime. Within the second section, I then applied this context more specifically to early radiographical practices in Britain. Addressing the complex and largely non-standardised landscape before and during the War, I have argued that wartime allowed nurses to seize novel opportunities to utilise this relatively novel technology, and examined shifts in attitude towards female radiographical nurses within Britain. Finally, I have looked at the varieties of radiographical nursing available to British women during wartime, citing the significance of radiology as an interdisciplinary profession.

As has been much discussed in recent historical narratives on women and the war, however, this was not the end of the story. As the war ended and men began to rejoin the workforce, this often forced women out of their wartime roles.[107] Nurses could not always continue their radiographical work after the war; however, radiology as women's work, and as a feminist cause, continued to impact the medical profession. For example, the Marie Curie Hospital in London was established in 1929 as a 'feminist organisation' to support both female doctors and the development of radium therapy, particularly for cervical cancer – 'a woman's disease'.[108] This became a well-respected institution, described in the prestigious medical journal *The Lancet* as a 'fitting memorial' to Curie, and therefore demonstrating the recognition of its success within British medicine.[109] The First World War therefore presented an opportunity for British nurses to play a role in the dissemination of X-ray technology in Europe, and to shape the nursing profession and women's involvement with future radiological work.

[107] Brock, 2021, 593-594. Brooks and Hallett, 2015, 1. Fara, 2018, 23-26, 27-28. Jones, Martin and Wolf, 2021, 20. Kelly, 2021, 541.
[108] Moscucci, 2016, 160.
[109] 'The Marie Curie Hospital', *The Lancet*, December 15 1934, pp. 1351-1352.

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